

DECLARATION: "I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>. I have also completed the Online Tutorial on avoiding plagiarism 'Ready Steady Write', located at <http://tcd.ie/libguides.com/plagiarism/ready-steady-write>."

Global Development Patterns: An Interactive Visualization

1. Dataset Description and Motivation

This project uses the Gapminder global development dataset, which contains social and economic data for countries across multiple years. Each record represents a country in a given year and includes GDP per capita, life expectancy, population, continent, and year.

The dataset is complex due to the large number of countries, multiple quantitative variables, and strong variation in scale. For example, GDP per capita ranges from a few hundred to many thousands, while population varies from very small countries to those with over a billion people. These differences make it difficult to understand relationships using a single static visualization, motivating the use of interactive multi-view techniques.

Although Gapminder is widely used, this project focuses on an individually designed dashboard that explicitly addresses visual complexity through coordinated views and interaction. A Top-N population filter is included to reduce clutter while preserving a meaningful global overview.

2. Data Complexity and Analytical Tasks

The dataset presents several sources of complexity. It includes multiple quantitative variables such as GDP per capita, life expectancy, and population, along with categorical grouping by continent. The data also spans multiple years, introducing a temporal aspect that increases the size and dimensionality of the dataset.

Displaying many countries simultaneously increases the risk of overplotting, which can obscure patterns and make interpretation difficult. Differences in scale between variables further complicate direct comparison, particularly when using static charts.

The visualization is designed to support a range of analytical tasks. We can compare countries based on economic and health indicators, explore the relationship between GDP per capita and life expectancy, and identify regional patterns across continents. The dashboard also supports multivariate analysis by allowing population size to be considered alongside economic and health measures.

The dashboard supports common analytical questions such as finding countries with similar development levels, comparing regions at different stages of development, and understanding how population size relates to economic and health outcomes. By using visualization, the dashboard makes these comparisons and patterns easier to explore and understand.

3. Visualization Design and Idioms Used

To handle the complexity of the dataset, an interactive coordinated multi-view dashboard was built using Plotly Dash. It combines multiple visual views in one interface, allowing users to explore different aspects of the data together.

The scatter plot serves as the primary view. It displays GDP per capita on a logarithmic scale against life expectancy, with point size representing population and colour representing continent. This view has direct comparison between countries and highlights the overall relationship between economic development and health outcomes.

A geographic map is included to provide spatial context for the selected countries and to support regional analysis. The parallel coordinates plot allows us to compare multiple quantitative variables simultaneously, making multivariate patterns easier to observe. Histogram shows the distribution of GDP per capita for the currently selected subset of countries.

These visual views are linked and work together. When we interact with one view, such as using lasso selection in the scatter plot, the same selection is shown in all other views. This coordination keeps the information consistent and allows users to explore the data without moving between separate or disconnected visualizations. Splitting information across multiple views also helps reduce the amount of information shown in any single view, making the dashboard easier to understand.

4. Interaction, Coordination, and Complexity Control

Interaction plays a central role in managing visual complexity within the dashboard. The main interaction technique is lasso selection in the scatter plot, which allows users to directly select subsets of countries based on their position in the data space.

Once a selection is made, all other views update automatically. The geographic map, parallel coordinates plot, distribution view, and KPI summaries reflect only the selected countries. This coordinated updating allows us to focus on specific subsets while maintaining a consistent analytical context.

Additional interaction is provided through a year slider, which allows users to explore different snapshot years and supports temporal analysis. A continent filter enables focused regional exploration by limiting the data to specific geographic groups. Furthermore, a Top-N population slider acts as an explicit complexity control by limiting the number of countries displayed, reducing overplotting and visual clutter. This interaction design follows the idea of showing an overview first, then allowing users to zoom in and filter the data as needed. Users can start with a global view and gradually narrow their focus through interaction, which helps manage complexity without overwhelming them.

5. Encoding Channels and Design Decisions

Several visual encoding channels were used to support clear analysis. Position encodes GDP per capita and life expectancy in the scatter plot, enabling accurate quantitative comparison. Colour is used consistently across all views to represent the continent, supporting visual consistency and rapid identification of regional groupings.

Size is used to encode population, highlighting differences in demographic scale between countries. Rather than using traditional faceting, the dashboard relies on coordination across multiple views, with each view acting as a functional facet that shows a different aspect of the same data.

A logarithmic scale is applied to GDP per capita to handle the highly skewed distribution of values and to

make patterns across low and high income countries easier to interpret. The dashboard is designed to fit within a single screen without scrolling, ensuring a stable layout and maintaining visual coherence, in line with the assignment requirements.

6. Strengths and Limitations

Strength: Effective coordination of multiple visual idioms within a single dashboard. Interaction is used meaningfully to support comparison, exploration, and subset analysis. Visual complexity is addressed through filtering and scaling techniques, helping reduce overplotting and improve interpretability.

The dashboard also maintains a clear visual hierarchy and uses consistent encodings across all views, which supports ease of use and smooth transitions between different representations.

Limitations: It focuses on high level national indicators and does not capture within country inequality or finer grained social differences. Temporal exploration is snapshot based rather than animated, and some views, such as the parallel coordinates plot, may require brief familiarisation for less experienced users.

Despite these limitations, the visualization demonstrates how interactive, coordinated multi-view design can effectively reduce complexity and support meaningful exploration of multivariate global data. Future extensions could include additional indicators or richer temporal interaction.

7. Tools and Technologies

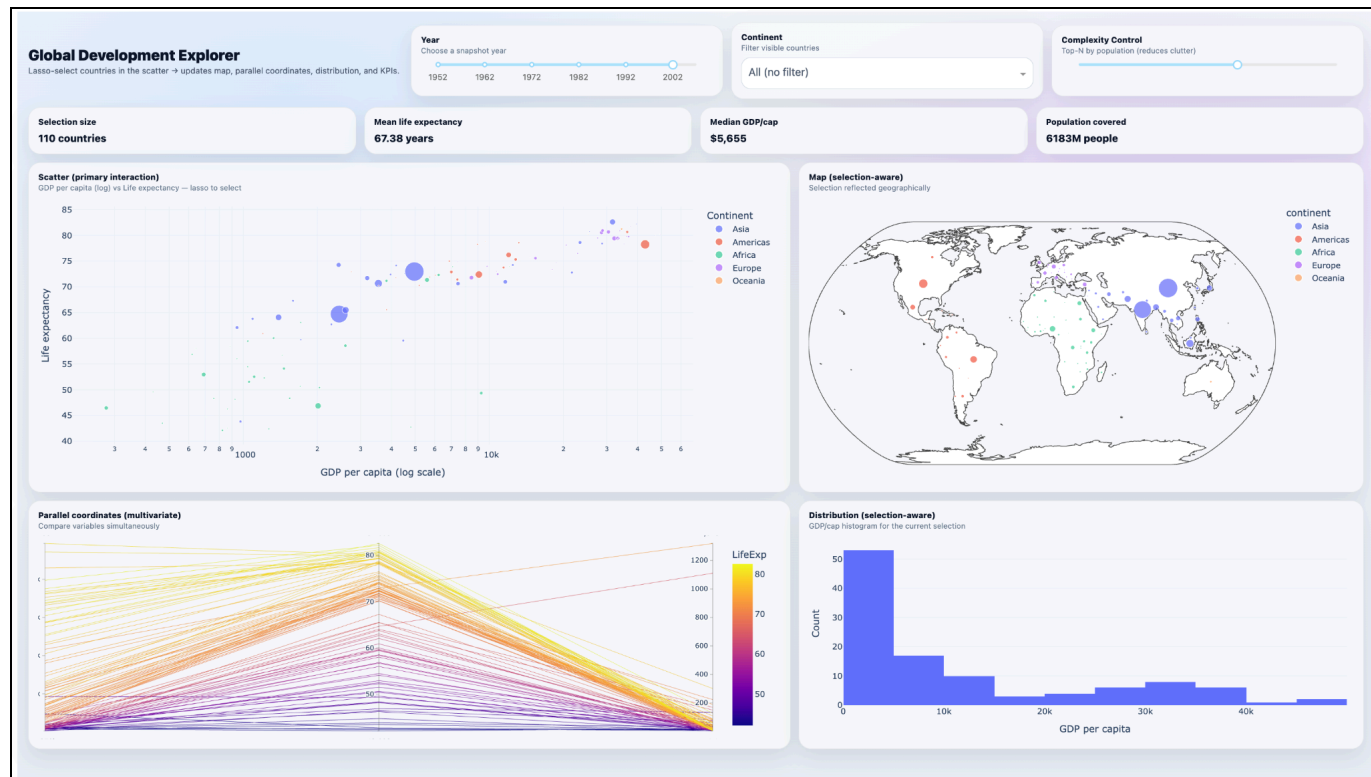
The visualization was implemented using Python and Plotly Dash, with Plotly Express used for chart creation and Pandas for data handling. These tools were chosen for their strong support for interaction, coordination, and rapid development of complex visualizations.

8. Conclusion

This project demonstrates how complexity in a multivariate global dataset can be addressed through coordinated multiple views and interaction. By integrating scatter plots, geographic maps, parallel coordinates, and distribution views into a single dashboard, the visualization enables effective exploration and comparison of global development patterns while remaining clear and interpretable.

9. References

- [1] Gapminder Foundation. *Gapminder Data*. <https://www.gapminder.org/data/>
- [2] Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press.
- [3] Ware, C. (2013). *Information Visualization: Perception for Design*. Morgan Kaufmann.
- [4] Plotly Technologies Inc. *Plotly Dash Documentation*. <https://dash.plotly.com/>



Overview of the interactive dashboard showing coordinated multiple views for global development data